

ETIOLOGY AND PATHOGENESIS

The pathogenesis of Sweet syndrome may be multifactorial and remains to be definitively determined. A septic process is suggested by the accompanying fever and peripheral leukocytosis. Indeed, a bacterial infection may have a causative role because most patients with classic Sweet syndrome have a febrile upper respiratory tract infection or tonsillitis that precedes their skin lesions by 1 to 3 weeks. Also, Sweet syndrome patients with culture-confirmed and serology-confirmed *Yersinia enterocolitica* intestinal infection improve with systemic antibiotics.

Sweet syndrome may result from a hypersensitivity reaction to an eliciting bacterial, viral, or tumor antigen. The appearance, histopathology, and course of the skin lesions support this concept. In addition, the prompt response of both the symptoms and the lesions to corticosteroids further supports this hypothesis.

Leukotactic mechanisms, dermal dendrocytes, circulating autoantibodies, immune complexes, HLA serotypes, and cytokines have all been postulated to contribute to the pathogenesis of Sweet syndrome. Complement does not appear to be essential to the disease process. In some patients, antibodies to neutrophilic cytoplasmic antigens have been demonstrated; however, these are likely to represent an epiphenomenon.

On the other hand, cytokines—directly and/or indirectly—may have an etiologic role in the development of Sweet syndrome symptoms and lesions.

G-CSF, granulocyte-macrophage colony-stimulating factor, interferon- γ , interleukin-1, interleukin-3, interleukin-6, and interleukin-8 are potential cytokine candidates in the pathogenesis.

CLINICAL FINDINGS

History

Sweet syndrome patients may appear dramatically ill. The skin eruption is usually accompanied by fever and leukocytosis. However, the skin disease can follow the fever by several days to weeks or be concurrently present with the fever throughout the episode of the dermatosis. Arthralgia, general malaise, headache, and myalgia are other associated symptoms.

Acute Febrile Neutrophilic Dermatositis (Sweet Syndrome)

At a Glance

A constellation of symptoms and findings: the acute onset of fever, neutrophilia, erythematous, and tender skin lesions that typically show an upper dermal infiltrate of mature neutrophils.

Cardiovascular, central nervous system, gastrointestinal, hepatic, musculoskeletal, ocular, oral, otic, pulmonary, renal, and splenic organs can be involved.

Classic Sweet syndrome may be associated with infection of the upper respiratory and/or gastrointestinal tract, inflammatory bowel disease, and pregnancy.

However, classic Sweet syndrome also occurs in younger adults and children. The youngest reported patients were brothers who developed the dermatosis at 10 and 15 days of age.

Several investigators consider it appropriate to distinguish between the classic form and the malignancy-associated form of this disease because the onset or recurrence of many cases is temporally associated with the discovery or relapse of cancer. Malignancy-associated Sweet syndrome does not have a female predominance and is most often associated with acute myelogenous leukemia.

In Sweet syndrome patients with dermatosis-related solid tumors, carcinomas of the genitourinary organs, breast, and gastrointestinal tract are the most frequently occurring cancers.

A drug-induced variant of the dermatosis has also been described. It has most frequently been observed in association with the administration of granulocyte-colony stimulating factor (G-CSF). However, several other medications have also been implicated in eliciting drug-induced Sweet syndrome.